Telecom Network Operations Assistant - Evaluation Report

Evaluation scope

The project evaluates Synapt retrieval, the complete orchestrator, prediction behavior, and RAG quality.
Judges use the AICredits OpenAI-compatible API with google/gemini-3.1-flash-lite where configured.
Scores are project evaluation results, not production service-level guarantees.

RAGAS Synapt evaluation

Queries: 3; good threshold: 0.6; overall score: 0.6050517429045752; frontend qualification: True.
RAGAS Faithfulness: average 0.7040441176470589, count 3, good=True.
RAGAS Context Precision: average 0.44444444439999997, count 3, good=False.
RAGAS Context Relevance: average 0.6666666666666666, count 3, good=True.
Context window was bounded to three contexts and 3,500 characters per context to control judge latency.

DeepEval Synapt evaluation

Status: completed; queries: 3; usable queries: 3; average score: 0.8780701754385966.
Metrics cover answer relevancy, faithfulness, contextual relevance/precision, and procedure/retrieval behavior depending on the case.

Orchestrator and prediction evaluation

Orchestrator test cases: 3; metrics: 9; passed metrics: 9; failed metrics: 0.
Prediction evaluation cases: 3; passed: 3; metrics: ["{'query': 'healthy_telemetry', 'metric': 'Prediction Agent Contract', 'score': 1.0, 'success': True, 'reason': 'Both model outputs are finite and satisfy the shared-input response contract.'}", "{'query': 'moderate_telemetry', 'metric': 'Prediction Agent Contract', 'score': 1.0, 'success': True, 'reason': 'Both model outputs are finite and satisfy the shared-input response contract.'}", "{'query': 'degraded_telemetry', 'metric': 'Prediction Agent Contract', 'score': 1.0, 'success': True, 'reason': 'Both model outputs are finite and satisfy the shared-input response contract.'}"].
The evaluation page reads saved reports from evaluations/reports and frontend/src/data.

Interpretation and limitations

Faithfulness measures support from retrieved context; contextual precision measures how focused the retrieved context is.
The moderate Context Precision score indicates retrieval noise or broad SOP/shared context and is an improvement area, not proof that every answer is incorrect.
AICredits/judge timeouts can make a metric unavailable; the reports record missing metrics and qualification logic.